

Erdős Problem #929

EP-929

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 2$ be large and let $S(k)$ be the minimal x such that there is a positive density set of n where $n+1, n+2, \dots, n+k$ are all divisible by primes $\leq x$.

Estimate $S(k)$ - in particular, is it true that $S(k) \geq k^{1-o(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>